Papagari Sumanth Reddy

em24mtech11005@iith.ac.in | sumanth9975@gmail.com | +91 9494987014 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.50
Jawahar Navodaya Vidyalaya, Wargal	Class XII (90.4)	2020	90.40%
Jawahar Navodaya Vidyalaya, Wargal	Class X (88.6)	2018	88.60%

SCHOLASTIC ACHIEVEMENTS

Ranked in the **top 5%** of 123,000 candidates nationwide

EXPERIENCE

Gen AI / Operations Research Intern -- Greenko Group

Designed and benchmarked warm-start strategies for a **Mixed-Integer Programming** solver powering Greenko's Day-Ahead energy-market scheduling, **cutting solver runtime by up to 71%** and winning 75% of benchmark cases against cold-start baselines (Python, Gurobi, Ray)

Engineered a production **GenAI agent** automating ISO/IMS documentation across 5 specialized pipelines -- LLM-based restructuring (**AWS Bedrock**), an **agentic** search assistant (self-hosted **vLLM**/Qwen), and native Word flowchart generation -- deployed with Keycloak SSO and Dockerized microservices

PROJECTS

KathaMitra -- Offline AI Storytelling Platform (M.Tech Thesis, IIT Hyderabad)

Designed a fully on-device pedagogical Android app using a quantized **Gemma 2B** LLM and local **vector search** (sqlite-vec), including a novel **Symmetric Context** retrieval framework to handle code-switched Hindi-English speech

Achieved under **450ms end-to-end latency** with **100% local inference**, validated through focus-group testing with children aged 9-12 that yielded a **4.80/5.0 pedagogical score**

Scientific Diagram Captioning -- Deep Learning & Computer Vision

Engineered a multimodal diagram-captioning pipeline, combining **CLIP** and **Vision Transformers** for visual feature extraction with **BART** for caption generation

Trained and evaluated the pipeline on a 4-million-image subset of the **SciCap** dataset, achieving **73% Recall@10** for caption-to-diagram retrieval

Multi-Feedstock Biogas Supply Chain Optimization -- Operations Research

Modeled multi-period sourcing and blending of four waste feedstocks across four Telangana biogas plants, accounting for a full year of seasonal variation in feedstock availability

Applied **prescriptive analytics** to maximize profitability while maintaining an optimal **20:1-35:1 C/N ratio** required for biogas yield

SKILLS

Languages & Data: Python, C++, SQL (MySQL, PostgreSQL), MongoDB

Machine Learning & Deep Learning: Regression, Classification, Clustering, CNN, RNN/LSTM, Transformers, Vision Transformers

GenAI & Agentic Systems: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), vLLM, AWS Bedrock, Prompt Engineering

Optimization & Backend: Gurobi, Mixed-Integer Programming, FastAPI, Docker, Ray, AWS

RELEVANT COURSEWORK

Natural Language Processing - Computer Vision - Deep Learning - Machine Learning - Operations Management - Strategic Management

POSITIONS OF RESPONSIBILITY

Hospitality Coordinator, SPIC MACAY IIT Hyderabad Chapter -- coordinated logistics for the 10th International Convention, hosting delegates and performers across multi-day cultural programming

EXTRACURRICULAR

Third Place, International Yoga Day Competition, IIT Hyderabad - Represented IIT Hyderabad in international research exchanges at Nagoya, Kyoto, and Shimane Universities (Japan)